

A Catalogue of Computational Methods for Single-Cell-Resolution Spatial Omics

TL;DR

The single-cell-resolution imaging-ST analysis stack now spans 14 task categories with 150+ methods; the **crowded** categories are spatial-domain/representation learning, cell-cell communication, and cell segmentation, while **sparse** high-opportunity categories are transcript decontamination, spatial RNA velocity, and QC/simulation for imaging ST specifically.

Community defaults in practice: **Xenium** → 10x multimodal segmentation or Proseg/Baysor + RCTD/SingleR + BANKSY/Squidpy; **CosMx** → AtoMx + InSituType + squidpy/Giotto; **MERFISH** → Baysor/Cellpose + scanpy/squidpy + BANKSY/CellCharter.

Many "spatial" methods are Visium-spot-native (RCTD, cell2location, SpatialDE, PASTE, iStar/TESLA, SpotClean) and only partially transferable; truly single-cell-native methods (Baysor, Proseg, InSituType, BANKSY, CellCharter, SIMVI, COMMOT, segger, ovrIpy/resolVI/SPLIT) are the ones to prioritize for imaging platforms.

Key Findings

This catalogue is organized by analysis task. For each method I give a one-line description/algorithm, year and venue, language/package, platforms, scalability where reported, and whether it is single-cell-native vs. spot-adapted. Preprints are flagged as not peer-reviewed.

A recurring theme directly relevant to a contamination-aware, disentangled, multi-task representation learner: transcript spillover/misassignment is now recognized as the central noise source in imaging ST (Xenium especially), and a distinct family of decontamination methods (ovrIpy, resolVI, SPLIT, DenoIST, CellAdmix) emerged in 2025–2026. Principled disentanglement of intrinsic vs. spatial/niche-induced variation is pioneered by SIMVI (with identifiability theory) and COVET/ENVI. No single method currently unifies decontamination + disentanglement + multi-task annotation/niche inference — this is the clearest white space.

Details

1. Cell segmentation (image-based)

Method	Description / algorithm	Year / venue	Impl.	Platforms
Cellpose 1/2/3	Generalist flow-field CNN; v2 human-in-loop training; v3 ("Cellpose3: one-click image restoration," Stringer & Pachitariu, Nat Methods 2025;22(3):592–599) nih ↗ adds denoise/deblur/upsample before segmentation	2021 Nat Methods; 2022 Nat Methods (2.0); 2025 Nat Methods (3.0)	Python (cellpose); models CC-BY-NC GitHub ↗	Any fluorescence/DAPI; used on Xenium/MERFISH nuclei
Cellpose-SAM	SAM-backbone generalization; Cellpose4 adds DINOv3 models	2025 bioRxiv (preprint)	Python	Multiplexed/fluorescence
StarDist	Star-convex polygon nucleus segmentation	2018/2020	Python (stardist)	DAPI nuclei; histology
Mesmer / DeepCell	ResNet50-FPN whole-cell+nuclear; trained on TissueNet (>1M cells, six platforms: CODEX/cyIF/IMC/MIBI/Vectra/MxIF)	2022 Nat Biotech	Python (deepcell)	Multiplexed protein imaging — de facto default for CODEX/IMC/MIBI
CellProfiler	Classic watershed/pipeline; paired with Ilastik for IMC	2021 (v4)	Python	IMC/general
ilastik	Random-forest pixel classification	2019 Nat Methods	Python/GUI	IMC segmentation pipeline (Bodenmiller)
InstanSeg	Embedding-based instance segmentation, fast	2024 (preprint/paper)	Python	Fluorescence/histology
CelloType	Transformer end-to-end detection+segmentation+classification	2024/2025 Nat Methods	Python	Multiplexed/spatial omics
CellSAM	SAM foundation model for cell segmentation	2023–2025 bioRxiv (preprint)	Python	Multiplexed/fluorescence
Vendor: 10x Xenium multimodal	DL nuclei + membrane expansion, pretrained on Xenium (FF/FFPE)	2023–	Xenium Ranger	Xenium

Method	Description / algorithm	Year / venue	Impl.	Platforms
Vendor: CosMx AtoMx / Bruker multimodal	Cloud GUI multimodal segmentation	2022-	AtoMx	CosMx
Vendor: MERSCOPE pipeline	Cellpose/watershed-based vendor	2021-	Vizgen	MERFISH

Note: For spatial proteomics (CODEX/IMC/MIBI), Mesmer/DeepCell (TissueNet-trained) is the confirmed default; alternatives include Cellpose, CellSeg (Mask R-CNN, for CODEX) and Dice-XMBD (U-Net for IMC). The Bodenmiller Ilastik+CellProfiler pipeline (now succeeded by steinbock) is the community IMC standard.

2. Transcript-aware / segmentation-free cell assignment

Method	Description / algorithm	Year / venue	Impl.	Platforms	Scale
Baysor	MRF over transcript point cloud; optional DAPI/scRNA priors; models 3D	2022 Nat Biotech	Julia	MERFISH/Xenium/ISS	limited >5mm²/>100 genes
Proseg	Cell simulation via cellular Potts model, probabilistic, 3D; models spillover; "Cell simulation as cell segmentation," Jones et al., Nat Methods 2025 — superior performance + efficiency benchmarked on MERSCOPE, CosMx and Xenium	2025 Nat Methods (DOI 10.1038/s41592-025-02697-0)	Julia/CLI	Xenium/CosMx/MERFISH	best avg across tissues
BIDCell	Self-supervised deep learning + biological loss functions; extends Cellpose	2024 Nat Commun	Python	Xenium/CosMx	large
segger	Heterogeneous GNN; transcript-to-cell link prediction; uses scRNA ref	2025 bioRxiv (preprint)	Python	Xenium (breast/lung/colon)	orders of magnitude faster than Baysor/BIDC

Method	Description / algorithm	Year / venue	Impl.	Platforms	Scale
					bioRxiv ↗
ComSeg	Point-cloud graph, no cell-shape prior Nature ↗	2024 Commun Biol	Python	MERFISH/ISS-type	complex tissue
JSTA	Joint segmentation + cell typing, DNN	2021 Nat Commun	Python	MERFISH	—
SCS	Sub-cellular segmentation, transformer + DL	2023 Nat Methods	Python	Stereo-seq/seq-based hi-res	—
Bering	GNN for molecule-level segmentation+annotation (preprint)	2023-2024	Python	imaging ST	—
ClusterMap	Density-based clustering of transcripts	2021 Nat Commun	Python	STARmap/MERFISH	—
SSAM / SSAM-lite	KDE-based segmentation-free cell typing	2021 Nat Commun	Python	osmFISH/MERFISH	—
SPArrOW (Sparrow)	Library for imaging ST analysis incl. segmentation	2024-	Python	imaging ST	—
Points2Regions	Fast clustering of categorical points; TissUUmaps/napari plugin	2024 Cytometry A	Python	ISS/MERFISH	multi-scale
FIGURE	Segmentation-free multilayered-Dirichlet pixel factorization (stochastic VI); handles billions of submicron points	2024 Nat Methods	Python	Stereo-seq/Seq-Scope/imaging	billions of points
Sainsc	Segmentation-free in-situ capture analysis	2024/2025	Python/R	Stereo-seq/imaging	scalable
GeneSegNet	DL joint gene+image segmentation	2023 Genome Biol	Python	MERFISH/ISS	—

Method	Description / algorithm	Year / venue	Impl.	Platforms	Scale
Nuc2Seg	Nucleus-to-cell expansion learning	2024-2025 (preprint)	Python	Xenium	—
TopACT	Topological single-cell classification of points	2024 (paper)	Python	Stereo-seq/MERFISH	—
Cellist	Combines imaging + expression, cross-platform	2026 Nat Genet	Python	multi-platform	scalable
UCS	Unified subcellular segmentation	2025 Small Methods	Python	subcellular ST	—
watershed / nuclear expansion	Baseline expansion of nuclei	—	various	all	Baseline

3. Segmentation error correction / decontamination / spillover & ambient

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
ovrlpy	Partitions transcripts into upper/lower z-plane subsets; flags regions of vertical signal mixing arXiv ↗	2026 (paper)	Python (ovrl.py)	Xenium/imaging	Native
resolVI	scVI-extension VAE; parameterizes contamination as latent variables; samples corrected counts biorxiv ↗	2025 (paper)	Python	Xenium/imaging	Native
SPLIT	Uses RCTD to detect pure vs. mixed cells; reweights counts by primary-type fraction; needs scRNA ref; biorxiv ↗ best cell-type separation + highest cosine similarity to Chromium in the Xenium benchmark Nature ↗	2026 Nat Methods (DOI 10.1038/s41592-026-03089-8)	R	Xenium	Native

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
DenoIST	Poisson mixture model; no reference; returns integer counts; bioRxiv ↗ outperforms resolVI/SPLIT on true-positive metric bioRxiv ↗	2025 bioRxiv (preprint)	—	imaging ST	Native
CellAdmix	Factorization of molecular-neighborhood graphs to detect/remove admixture molecules arXiv ↗	2026 (Kharchenko lab)	R	imaging ST	Native
FastReseg	Log-likelihood-ratio scoring per transcript; flags & reassigns mislocalized transcripts arXiv ↗	2024 (Danaher)	R	CosMx	Native
ENACT	End-to-end nuclei-to-cell assignment for Visium HD	2025	Python	Visium HD	Native (hi-res)
SpotClean	Corrects spot swapping — Visium spot-based only	2021 Nat Commun	R	Visium	Spot-only
SoupX / CellBender / DecontX	Ambient-RNA correction — designed for droplet scRNA-seq; adaptable	2020-2021	R/Python	scRNA (adapted)	Adapted

This is the sparsest, highest-value category for a contamination-aware model: five imaging-native methods, all 2025–2026, no established winner.

4. Cell type annotation / label transfer for targeted panels

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
RCTD / spacexr	Robust cell-type decomposition; reference-based; doublet/full modes	2021 Nat Biotech	R (spacexr)	designed for Visium/Slide-seq; validated on imaging	Spot-designed, works SC
InSituType	Likelihood-based Bayes/EM; supervised/unsupervised/semi; biorxiv ↗ leverages images + neighborhood; bioRxiv ↗	2022 bioRxiv (preprint)	R	CosMx (native; AtoMx default) NanoString ↗	Native

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
	escalating-subsample scheme biorxiv ↗				
Tangram	Deep-learning mapping of scRNA to space	2021 Nat Methods	Python	MERFISH/Visium	Adapted
CellTypist	Logistic-regression classifier	2022 Science	Python	scRNA (adapted to ST)	Adapted
SingleR	Correlation-based reference annotation; judged best reference-based tool for Xenium (Cheng, Jin, Smyth & Chen, BMC Bioinformatics 2025;26 — "fast, accurate and easy to use" vs Azimuth/RCTD/scPred/scmapCell) BMC Bioinformatics ↗	2019	R	scRNA (best for Xenium in benchmark)	Adapted
scANVI	Semi-supervised VAE annotation	2021	Python (scvi)	scRNA/ST	Adapted
Symphony	Reference-mapping compression	2021 Nat Commun	R	scRNA/ST	Adapted
TACCO	Optimal-transport annotation & composition	2023 Nat Biotech	Python	Visium/imaging	Both
SpaGE	KNN domain-adaptation imputation/annotation	2020	Python	MERFISH/Visium	Adapted
moscot	Optimal transport at scale for mapping	2024/2025 Nature	Python	imaging/scRNA	Both

5. Representation learning / embedding / spatial clustering

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
BANKSY	Neighborhood-augmented product space (own + neighbor transcriptome); unifies cell typing + domains; processes millions of	2024 Nat Genet	R/Python	imaging+seq RNA, protein	Native

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
CellCharter	cells Nature ↗				
	Neighborhood aggregation Nature ↗ + GMM; platform-agnostic; hundreds of samples / millions of cells Nature ↗	2024 Nat Genet	Python	CODEX/CosMx/imaging	Native
NicheCompass	Interpretable GNN with gene-program priors; multi-sample; among few methods handling >70k-cell datasets ResearchGate ↗	2025 Nat Genet	Python	MERFISH/CosMx/Xenium	Native
SIMVI	Annotation-free VAE disentangling cell-intrinsic vs. spatial-induced latents with identifiability theory; single-cell-resolution spatial effects (Dong, Su, Kluger et al., Nat Commun 2025;16:2990)	2025 Nat Commun	Python	MERFISH/Slide-seq/spatial multiome	Native (disentangled)
COVET / ENVI	Covariance-environment niche representation + conditional-VAE integration of scRNA & spatial nih ↗	2024 Nat Biotech	Python	MERFISH/imaging	Native
MENDER	Multi-range neighborhood aggregation; fast domains	2024 Nat Commun	Python	imaging ST	Native
UTAG	Linear neighbor-weighting;	2022 Nat Methods	Python	IMC/CODEX/imaging	Native (protein)

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
	unsupervised domains				
STAGATE	Graph-attention autoencoder	2022 Nat Commun	Python	Visium/imaging	Both
GraphST	Graph contrastive learning	2023 Nat Commun	Python	Visium/imaging	Both
SpaceFlow	GNN embedding + pseudo-spatial-map	2022 Nat Commun	Python	imaging/seq	Both
CCST	Graph-conv clustering	2022 Nat Comput Sci	Python	imaging/Visium	Both
SEDR / conST / DeepST / SpaGCN	GNN/autoencoder domain methods, largely Visium-first	2021-2023	Python	mostly Visium	Spot-first
SpatialPCA	Spatially-aware PCA	2023 Nat Commun	R	Visium/Slide-seq	Spot-first
BASS / PRECAST / SpiceMix	Multi-sample/factor spatial clustering	2022-2023	R/Python	Visium/MERFISH	Both
STAligner / SPACEL / stMSA	Integration/embedding across slices	2023-2024	Python	multi	Both
SpatialGlue / SpaMosaic	Multi-omics spatial integration	2024-2025 Nat Methods	Python	spatial multi-omics	Native
FM: Nicheformer	Transformer trained on SpatialCorpus-110M PubMed ↗ (57.06M dissociated + 53.83M spatial cells across 73 PubMed ↗ organs, four spatial technologies);	2025 Nat Methods	Python	Xenium/MERFISH/CosMx	Native

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
FM: Novae	PubMed ↗ Schaar/Tejada-Lapueta et al., Nat Methods 2025 (DOI 10.1038/s41592-025-02814-z)	2025 Nat Methods	Python (MICS-Lab)	Xenium/MERSCOPE/CosMx	Native
	Graph-attention SSL FM trained on 78 slides ≈ 29–30M cells across 18 tissues and three subcellular biorxiv ↗ technologies (Xenium, MERSCOPE, CosMx); zero-shot nested domains (Blampey et al., Nat Methods 2025)				
FM: scGPT-spatial	Continual-pretrained scGPT; MoE decoder; SpatialHuman30M arxiv ↗	2025 bioRxiv (preprint)	Python	Visium/HD/Xenium/MERFISH	Both
FM: CellPLM	Cells-as-tokens transformer w/ spatial positional encoding	2024 ICLR / preprint	Python	scRNA+ST	Both
FM: SToFM / STFormer / SpaFormer	Niche/gene/imputation FM variants	2024-2025 (preprints)	Python	ST	mixed

6. Niche / tissue domain / cellular neighborhood discovery

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
CellCharter, NicheCompass, MENDER, UTAG, BANKSY (domain mode)	see §5	—	—	—	Native

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
CytoCommunity	GCN on cell-type-label graphs + MinCut pooling; supervised & unsupervised; robust to expression noise (uses categorical labels)	2024 Nat Methods	Python	CODEX/MIBI/IMC	Native (protein)
Spatial-LDA	Latent Dirichlet allocation of neighborhood composition	2020	Python	CODEX/MIBI	Native (protein)
Schürch-style CN windows	k-nearest neighborhood composition + k-means	2020 Cell	—	CODEX	Native (protein)
MISTy	Multiview modeling of marker interactions/views	2022 Genome Biol	R	IMC/Visium/imaging	Both
spicyR	Spatial cross-type L-function statistics; multi-sample	2022 Bioinformatics	R	IMC/CODEX	Native (protein)
Kontextual	Relative cell-type localization inference	2024	R	IMC/CODEX	Native
NCEM	Node-centric expression GNN modeling communication	2022-2023 Nat Biotech	Python	imaging ST	Native
GraphCompass	Graph metrics for multi-sample niche comparison (scverse)	2024	Python	imaging ST	Native
scNiche	Neighborhood embedding	2024-2025	Python	imaging ST	Native

7. Cell-cell communication / ligand-receptor (spatial)

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
COMMOT	Collective optimal transport with spatial-distance constraints; competition among LR species; Nature ↗ validated on eight datasets / five technologies biorxiv ↗ incl. MERFISH/seqFISH	2023 Nat Methods	Python	imaging/Visium	Native

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
SpatialDM	Global/local bivariate Moran's R for LR co-expression hotspots	2023 Nat Commun	Python	imaging/Visium	Both
CellChat v2	Curated LR DB + communication probabilities; spatial support added	2023 Nat Protocols/updates	R	scRNA + spatial	Adapted
CellPhoneDB v3/v5	Statistical LR permutation; v3+ spatial microenvironments	2022-2024	Python	scRNA + spatial	Adapted
SpaTalk	Knowledge-graph LR + downstream target inference	2022 Nat Commun	R	imaging/Visium	Both
stMLnet	Multilayer signaling-regulation networks	2023	R	spatial	Both
HoloNet	GCN of combined communication events	2023 Nat Commun	Python	Visium/imaging	Both
NicheNet	Ligand-to-target gene regulatory potential	2020 Nat Methods	R	scRNA (adapted)	Adapted
SpaOTsc	Optimal-transport spatial communication	2020	Python	imaging	Native
scriabin / NICHES / Renoir / DeepLinc / SCOTIA	Various CCC frameworks; DeepLinc = de novo VAE interaction landscape	2022-2024	R/Python	mixed	mixed
NicheDE / CellNeighborEX	Niche-differential / neighbor-dependent DE for communication	2023-2024	R/Python	imaging/Visium	Both
LIANA+	Consensus/ensemble CCC framework	2024 Nat Cell Biol	Python	multi	Both
MEBOCOST	Metabolite-mediated communication	2022-2024 (preprint)	Python	scRNA/spatial	Adapted

8. Spatially variable genes / spatial statistics

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
SPARK-X	Non-parametric covariance test; linear scaling; top SVG-benchmark performer (avg corr 0.88) Springer ↗	2021 Genome Biol	R	Visium/Slide-seq/imaging	Both
nnSVG	Nearest-neighbor GP; per-gene length scale; bioRxiv ↗ linear scaling	2023 Nat Commun	R	Visium/Slide-seq/imaging	Both
SpatialDE / SpatialDE2	Gaussian-process spatial variance; DE2 quadratic scaling	2018/2021	Python	Visium (spot-first)	Spot-first
SPARK	GLSM Poisson/Gaussian; Nature ↗ cubic scaling Nature ↗	2020 Nat Methods	R	Visium	Spot-first
MERINGUE	Spatial autocorrelation w/ non-uniform density	2021 Genome Res	R	imaging/seq	Native
SpaGFT	Graph Fourier transform	2024	Python	Visium/imaging	Both
Moran's I / Geary's C (Squidpy)	Spatial-autocorrelation baselines	2022	Python	all	Both
SOMDE / Sepal / scGCO / singleCellHaystack	GP/diffusion/graph-cut SVG variants	2021-2022	Python	mixed	Both
ctSVG	Cell-type-specific SVGs Springer ↗ for Visium HD single-cell	2025 Genome Biol	R	Visium HD	Native (hi-res)

9. Trajectory / dynamics / RNA velocity with space

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
STT (spatial transition tensor)	4D transition tensor + spatial-constrained random walk;	2024 Nat Methods	Python (omicverse)	MERFISH/spatial	Native

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
	multistability & attractors				
TopoVelo	Spatially coupled ODEs extending RNA velocity to whole tissue	2025 Nat Biotech	Python	imaging ST (cortex)	Native
SIRV	Spatially resolved velocity; uses scRNA ref to enrich unspliced/spliced	2023	Python	imaging ST	Native
SpaceFlow pSM	Pseudo-spatial-map ordering	2022 Nat Commun	Python	imaging/seq	Native
SpaceOracle	Spatial GRN / perturbation modeling	2024-2025 (preprint)	Python	spatial	Native
Palantir / Monocle3 / PAGA / scFates / DPT	Post-embedding pseudotime (non-spatial, applied post hoc)	various	R/Python	any	Adapted

Sparse category — few truly spatial-native velocity methods exist.

10. Multi-slide / 3D alignment and registration

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
PASTE	Fused Gromov-Wasserstein OT alignment (full overlap)	2022 Nat Methods	Python	Visium/MERFISH	Both
PASTE2	Partial-FGW OT for partial overlap + 3D reconstruction	2023 (preprint/paper)	Python	Visium/Stereo-seq	Both
moscot	Scalable OT for mapping/alignment	2024/2025 Nature	Python	imaging/seq	Both
SLAT	Graph-adversarial heterogeneous alignment	2023 Nat Commun	Python	imaging/seq	Both
GPSA	Gaussian-process common coordinate	2022 NeurIPS/paper	Python	Visium/imaging	Both
CAST	Graph-based physical alignment at single-cell resolution	2024 Nat Methods	Python	MERFISH/imaging	Native

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
STalign	Diffeomorphic (LDDMM) image registration	2023 Nat Commun	Python	MERFISH/Xenium/Visium	Both
Eggplant	GP transfer to common reference	2023	Python	multi	Both
SPACEL (Scube)	Deep-learning 3D stacking/alignment	2023 Nat Commun	Python	imaging/Visium	Both
11. Multi-modal integration (transcripts + morphology/H&E/protein)					
Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
iStar	Hierarchical ViT (HIPT) histology features → super-resolution expression	2024 Nat Biotech	Python	Visium+H&E (spot super-res)	Spot-first
TESLA	Histology-guided super-resolution + TME annotation	2023 Cell Systems	Python	Visium+H&E	Spot-first
SpatialGlue	Dual-attention spatial multi-omics integration	2024 Nat Methods	Python	spatial multi-omics	Native
MaxFuse	Cross-modal weakly-linked matching (protein↔RNA)	2024 Nat Biotech	Python	CODEX↔scRNA	Native
SpaMosaic	Mosaic spatial multi-omics integration	2025	Python	spatial multi-omics	Native
CellSighter	CNN ensemble phenotyping on raw multiplexed images	2023 Nat Commun (Keren Lab)	Python	MIBI/IMC/CODEX	Native (protein)
Pathology FMs (UNI, Virchow, CONCH, Prov-GigaPath)	H&E foundation encoders applied to paired ST	2024-2025	Python	H&E + ST	Adapted
12. Differential expression / compositional / multi-sample statistics					
Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
C-SIDE (spacexr)	Cell-type-specific DE via additive log-linear mixture model; multi-replicate	2022 Nat Methods	R	Slide-seq/MERFISH/Visium	Both

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
	inference; validated on MERFISH/Slide-seq/Visium				
scCODA / tascCODA	Bayesian compositional analysis (+ tree)	2021-2022	Python	any	Adapted
milode / milo	KNN-neighborhood differential abundance/expression	2023-2024 Nat Biotech/preprint	R	scRNA/spatial	Adapted
sccomp	Compositional DA w/ variability modeling	2023 PNAS	R	any	Adapted
Hotspot	Informative gene modules across modalities	2021 Cell Systems	Python	scRNA/spatial	Both
distinct / PILOT	Distributional DE / patient-level ordering	2022-2023	R/Python	any	Adapted
cell2location	Spot deconvolution — Visium spot-based	2022 Nat Biotech	Python	Visium/spot	Spot-only

13. Quality control, simulation, benchmarking

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
SRTsim	Spatial-pattern-preserving simulator; only method consistently strong on spatial metrics in benchmarks	2023 Genome Biol	R	Visium/Slide-seq/imaging	Both
scDesign3	Unified probabilistic simulator for sc + spatial	2024 Nat Biotech	R	multi	Both
spaSim (SPIAT)	Simulates cell locations/tissue structure (no expression)	2023 Nat Commun	R	multiplexed	Native (protein)
scMultiSim	GRN + CCI-guided sc/spatial simulation	2024	R	spatial	Both
scSpatialSIM	Spatial single-cell molecular simulator	2025	R	multiplexed	Native
CRISP	Purity metric (co-expression rejection in segmentation) for segmentation benchmarking, no ground truth ResearchGate ↗	2026 bioRxiv (preprint)	R/Python	Xenium (5k panel)	Native

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
Independent benchmarks	Xenium cell-typing (2025), niche/domain-segmentation (2026), SVG (2025), segmentation/decontamination (2025-2026)	2024-2026 —		Xenium/imaging	—
Squidpy/scanpy QC	Standard QC workflows	2022	Python	all	Both

14. Data structures, infrastructure, visualization

Method	Description / algorithm	Year / venue	Impl.	Platforms	Native?
SpatialData (scverse)	Universal framework; OME-NGFF Zarr (images/labels/tables), Parquet (shapes/points); interoperates with scanpy/squidpy/scvi-tools	2025 Nat Methods	Python	all imaging+seq	N
AnnData / MuData	Core single-cell / multi-omics containers	2024 JOSS	Python	all	B
Squidpy	Spatial single-cell toolkit; neighborhood graph, spatial stats, image layer; Readthedocs ↗ ~10× faster than Giotto for some ops	2022 Nat Methods	Python	all	N
Giotto Suite	Technology-agnostic multi-scale multi-omics R ecosystem biorxiv ↗	2021 / 2025 (Suite)	R	all	N

Method	Description / algorithm	Year / venue	Impl.	Platforms	Notes
Seurat v5 spatial	Spatial workflows in Seurat	2023-2024	R	Visium/imaging	Bo
SpatialExperiment / SpatialFeatureExperiment	Bioconductor containers	2022-2023	R	all	Bo
Voyager	ESDA / geospatial spatial statistics (Pachter lab)	2023-2024	R/Python	all	Na
Bento	Subcellular transcript analysis toolkit	2024 Genome Biol	Python	imaging ST	Na
MoleculeExperiment	Molecule-resolved Bioconductor infrastructure	2023 Bioinformatics	R	imaging ST	Na
SOPA	Technology-invariant pipeline for image-based spatial omics	2024 Nat Commun	Python	Xenium/MERSCOPE/CosMx	Na
TissUUmaps	Fast web point/image viewer	2022-2023	JS/Python	imaging ST	Na
Vitessce	Web integrative viewer	2021-2024	JS	all	Na
napari-spatialdata	Interactive napari plugin	2024-2025	Python	all	Na
Vendor viewers: Xenium Explorer, AtoMx	Interactive vendor GUIs	—	GUI	Xenium/CosMx	Na
Spatial-proteomics-specific phenotyping (CODEX/PhenoCycler, IMC, MIBI-TOF)					
Method	Description / algorithm	Year / venue	Impl.	Platforms	
Astir	Bayesian marker-based cell-type assignment; deep recognition net for fast variational inference; expression-only	2021 Cell Systems	Python (camlab)	IMC/CyTOF/CyCIF (~2M cells)	

Method	Description / algorithm	Year / venue	Impl.	Platforms
STELLAR	Geometric deep learning / GCN on spatial kNN graph; transfers annotations + discovers novel types	2022 Nat Methods	Python (snap-stanford)	CODEX (intestine)
CELESTA	Unsupervised MRF optimization; anchor cells + spatial neighbor score	2022 Nat Methods	R (plevritis-lab)	CODEX (CRC/HNSCC)
Pixie (ark-analysis)	SOM pixel clustering → cell clustering; robust incl. ECM	2023 Nat Commun	Python (angelolab)	MIBI-TOF
MAPS	MLP deep classifier; pathologist-level annotation	2023-2024 Nat Commun	Python	MIBI/CODEX
STARLING	Segmentation-aware probabilistic clustering that models doublets/mixed profiles	2025 Nat Commun	Python (camlab)	IMC (gold-standard pellet)
Nimbus	DL per-cell marker-positivity prediction across any panel; trained on Pan-Multiplex (197M annotations)	2025 Nat Methods (bioRxiv 2024)	Python (angelolab)	MIBI/CODEX/IMC
MAUI	MATLAB GUIs for spillover/denoise — pre-processing, NOT phenotyping	2021 PLoS Comput Biol	MATLAB	MIBI/IMC

Recommendations

Staged next steps for building a contamination-aware, disentangled, multi-task representation learner:

Anchor on the decontamination frontier first (sparse, high-value). Your direct baselines are ovrIpy, resolVI, SPLIT, DenoIST and CellAdmix — all 2025–2026, all imaging-native, with no established winner. Benchmark using the SPLIT paper’s Xenium framework (Nat Methods 2026) plus the CRISP purity metric and DenoIST’s PMP/MECR metrics. Threshold to beat: match SPLIT’s cell-type separation and cosine similarity to Chromium snRNA-seq while **retaining more cells than ovrIpy/resolVI**, which empty out many cells after QC — this is their documented weakness and your clearest differentiator.

Position disentanglement against SIMVI and COVET/ENVI. SIMVI (Nat Commun 2025) is the only method with identifiability theory for intrinsic-vs-spatial disentanglement; ENVI adds transcriptome-wide imputation. A multi-task model that jointly performs decontamination + disentanglement + niche/annotation would occupy genuine white space, since no current tool unifies these. [biorxiv](#)

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[Nature](#)

Standardize preprocessing on per-platform defaults: Xenium → Proseg (best-benchmarked) or 10x multimodal; CosMx → InSituType after AtoMx; MERFISH → Baysor/Cellpose. Build on SpatialData/AnnData containers and SOPA for pipeline plumbing so the method is adoptable.

Prove scalability at millions of cells. BANKSY, CellCharter, Novae (~30M) and Nicheformer (~54M spatial) all operate at this scale. A method that cannot handle >1M cells will not be adopted for Xenium 5k/whole-brain atlases — treat 1M cells as the minimum viable scale and 10M+ as competitive.

Only invoke foundation-model comparison if claiming generality. Nicheformer and Novae are the peer-reviewed spatial FMs to benchmark against; scGPT-spatial and CellPLM are preprint/emerging and can be cited as related work.

Benchmarks that would change these recommendations: a new independent decontamination benchmark that names a clear winner; a foundation model demonstrating zero-shot decontamination; or a published unified multi-task method that supersedes current single-task tools (watch NicheCompass, SIMVI and Novae follow-ups).

Caveats

Preprints, not peer-reviewed (verify status before citing): Cellpose-SAM, segger, DenoIST, scGPT-spatial, CellPLM, CRISP, and the original InSituType bioRxiv (2022) — though InSituType is now the deployed default in Bruker AtoMx. Several 2026-dated items (ovrlpy, CellAdmix) are at the preprint/early-publication boundary; SPLIT and Proseg are confirmed in Nature Methods.

Spot-only methods flagged as not directly applicable to single-cell imaging without adaptation: SpotClean, cell2location, and largely SpatialDE/SPARK, iStar, TESLA. **RCTD and C-SIDE** were designed for spot/bead data but are validated on imaging platforms (MERFISH), so they occupy a middle ground.

"Native vs. adapted" is my judgment from demonstrated platforms and algorithmic assumptions; many "Both" methods run at single-cell resolution but were spot-first in design and may carry spot-scale assumptions (e.g., mixture/deconvolution priors) that are unnecessary or mis-specified at true single-cell resolution.

Exact year/venue/DOI for a few rapidly-evolving tools (Cellist, CellAdmix, ovrlpy, some FM variants) should be reconfirmed against the latest published versions.

Attribution correction worth noting: CellSighter is a Keren Lab (Weizmann) method; MAUI is a pre-processing/denoising tool, not a phenotyping method; Mesmer/DeepCell and the Ilastik pipeline are segmentation (not annotation) steps.

This catalogue is extensive but not exhaustive — representation learning and cell-cell communication in particular produce new methods monthly.